MOTILAL NEHRU NATIONAL INSTITUTE OF TECHNOLOGY
Data Structures Lab

Assignment 09

1. Implement a binary tree, supporting the following operations:

    a) **Insert()**: Insert a node into the tree.
    b) **Display()**: Display the tree using *in-order traversal*.

2. Implement a binary search tree (BST) with the following operations:

    a) **Insert()**: Insert a node in the BST.
    b) **Search()**: Search for a given element in the BST and return whether it exists.
    c) **Delete()**: Delete a node from the BST.

3. Implement the three types of depth-first traversal methods in a binary tree:

    a) **In-order Traversal**: Left -> Root -> Right
    b) **Pre-order Traversal**: Root -> Left -> Right
    c) **Post-order Traversal**: Left -> Right -> Root

4. Write a C program to calculate the height of a binary tree. Ensure the program handles empty trees situation.

5. Write a C program to perform **level-order traversal** (breadth-first traversal) of a binary tree. In level-order traversal, nodes are visited level by level from left to right.